

LARGE LANGUAGE MODELS

How ChatGPT Works [cite: 616]

The Super-Predictor: A Guide for Grade 6 [cite: 617]

Planrs Academy | Session 1

Unlocking the Patterns of Human Language

What is ChatGPT? [cite: 616, 617]

ChatGPT is not a robot or a person. [cite: 617] It's a **super-smart predictor** that guesses what word comes next! [cite: 617]

A Clever Helper: Think of it as a clever helper that has read millions of sentences. [cite: 618] It uses those sentences to predict words that help you write and learn! [cite: 619]

It doesn't "think" like a human—it just sees patterns in language! [cite: 622]

Understanding LLMs [cite: 620]

A **Large Language Model (LLM)** is a computer program that understands and writes text by predicting the next word! [cite: 620, 621]

It's like a librarian who has read every book in every school library in India. [cite: 628] Because they've read so much, they can guess exactly what word should follow another. [cite: 629]

Keywords to Remember: [cite: 621]

- **Predict:** Guessing the next part of a sentence.
- **Pattern:** Seeing how words usually go together.
- **Library:** The massive collection of data the AI learned from.

Learning Through Patterns [cite: 622]

Just like in **Antakshari**, where one word or sound leads to another, an LLM learns patterns by reading millions of sentences. [cite: 622]

Pattern Recognition: When you sing a song ending with "na," the next person starts with "na." [cite: 623] The AI has read so many sentences that it knows exactly which words follow others! [cite: 624]

It's like playing a giant, never-ending game of word-matching! [cite: 624]

A Super-Powered Smartphone [cite: 625]

Have you seen how your phone suggests the next word when you type? ChatGPT is like a **super-powered version** of that! [cite: 625]

Example: Type "Happy..." and your phone might suggest "Diwali" or "Birthday." [cite: 626] LLMs predict words much faster and smarter than any phone because they've seen millions of examples! [cite: 627]

It's all about guessing what's most likely to come next! [cite: 617]

The Digital Librarian with a Global Brain [cite: 628]

Imagine a librarian who has read every book in every school in India. [cite: 628]
They can guess what words come next in sentences about Indian History or Science! [cite: 629]

The Brain of Books: ChatGPT is like this super-librarian. [cite: 630] It uses patterns from millions of texts to predict helpful answers for you! [cite: 631]

It doesn't "know" facts the way you do; it just knows which words look right together! [cite: 631]

The Giant Digital Library [cite: 632, 633]

Where does ChatGPT learn from? It reads **billions of sentences** to understand patterns in language! [cite: 633]

Training Data Includes: [cite: 634-636]

- Sentences from books and newspapers like **The Hindu**.
- Indian English phrases and names of foods like **Samosa**.
- Stories about heroes like **Mahatma Gandhi** and **Rani Lakshmibai**.

Next-Word Prediction Game [cite: 637]

Let's play a guessing game! Can you predict what word comes next in these famous Indian phrases? [cite: 638]

"Mera Bharat ____" [cite: 639]

"Satyameva ____" [cite: 639]

You're using pattern recognition, just like ChatGPT! [cite: 639]

Indian Phrase Examples [cite: 639]

See how pattern recognition works with phrases you already know! [cite: 639]

- **"Mera Bharat Mahan"** (My India is Great)
- **"Satyameva Jayate"** (Truth Alone Triumphs)

Because you've heard these many times, your brain predicts the next word instantly. [cite: 638, 639] AI does the same thing! [cite: 639]

Writing Clear Instructions [cite: 640]

You are the "**Prompt Engineer**"! [cite: 640] The clearer your instructions, the better the AI's answers will be. [cite: 641]

Ask in Simple Steps: Think about what you want to know and ask in specific, simple steps. [cite: 642]

Example: "Tell me how to make Masala Chai in 3 simple steps." [cite: 643]

Your Turn to Prompt! [cite: 644]

Try writing these prompts to an AI and see what creative responses you get!
[cite: 644]

- "Tell me a fun fact about Mahatma Gandhi." [cite: 645]
- "Write a 2-line poem about a Rainy Day in India."
- "Give me 5 items needed for a Picnic at the beach."

The better your prompt, the better the prediction! [cite: 641]

What Did We Learn Today? [cite: 645]

- **ChatGPT is a Predictor:** It guesses the next word based on patterns! [cite: 646]
- **Learning from Data:** It learns by reading billions of sentences from books and websites. [cite: 646]
- **Prompts Matter:** Clear instructions help the AI give useful answers. [cite: 647]

The Human in the Loop [cite: 647]

AI is a powerful tool, but it's not always right. [cite: 647] Always remember that **you** are the smart one!

Stay Curious: Ask questions, explore new ideas, and use AI to help you learn, but always double-check the facts! [cite: 647]

Now you know how ChatGPT predicts words just like magic! [cite: 648]

THANK YOU! [cite: 647]
Stay Curious & Keep Learning! [cite: 647]

The best answers come from asking great questions. [cite: 649]

Planrs Academy | Empowering Grade 6